

UNIT – I

COMPUTER FUNDAMENTALS

COMPUTER:

- Common Operating Machine Purposely Used for Technological and Educational Research
- Charles Babbage, an English mechanical engineer, considered as the "Father of the Computer", he conceptualized and invented the first mechanical computer in the early 19th century.

Definition :

- The word Computer comes from the word “compute”, which means, “to calculate”.
- A computer is an electronic device that processes data according to instructions provided by computer programmers.
- A computer is an electronic device that can perform arithmetic operations at high speed.
- A computer is also called a data processor because it can store, process, and retrieve data whenever desired.
- It can perform a wide range of tasks, from basic calculations to complex operations.

HISTORY AND EVOLUTION OF COMPUTER

- The first counting device was used by the primitive people.
- They used sticks, stones and bones as counting tools.
- As human mind and technology improved with time more computing devices were developed.
- Some of the popular computing devices starting with the first to recent ones are History of Computer.

1. ABACUS :

- The History of computer begins with birth of abacus which is believed to be the first computer.
- Abacus was invented by the Chinese around 4000 years ago.
- Abacus is a wooden rack with metal rods with beads attached to them.
- The abacus operator moves the beads according to certain guidelines to complete arithmetic computations.
- Beads represents digits , Rods represents places – units, tens, hundreds and higher multiples of ten.
- It was primarily used for addition, subtraction , multiplication and division.
- It can also be used to extract square roots & cube roots of a number.

2. NAPIER'S BONES :

- It was a manually-operated calculating device which was invented by John Napier (1550-1617) of Merchiston.
- In this calculating tool, he used 9 different ivory strips or bones marked with numbers to multiply and divide.
- So, the tool became known as "Napier's Bones.
- It was also the first machine to use the decimal point.
- It has a set of rods or bones with a multiplication tables on them.
- Performed addition, subtraction, multiplication and division.
- The left edge had 9 numbers from 1 to 9.
- Allowed operator to multiply, divide and calculate square and cube roots by moving the rods around & placing them in specially constructed boards.

3. PASCALINE:

- Pascaline is also known as Arithmetic Machine or Adding Machine.
- It was invented between 1642 and 1644 by a French mathematician- philosopher Blaise Pascal.
- It is believed that it was the first mechanical and automatic calculator.
- It could only perform addition and subtraction.
- It was a wooden box with a series of gears and wheels. When a wheel is rotated one revolution, it rotates the neighboring wheel.
- It was a rectangular box with 8 movable wheels.
- A series of windows is given on the top of the wheels to read the totals.
- Addition & Subtraction was faster than abacus.
- This could also do the carry over operation.

4. STEPPED RECKONER OR LEIBNITZ WHEEL :

- It was developed by a German mathematician-philosopher Gottfried Wilhelm Leibnitz in 1673.
- He improved Pascal's invention to develop this machine.
- It was a digital mechanical calculator which was called as the Stepped Reckoner as instead of gears it was made up of fluted drums.
- It performs only Multiplication and Division.

5. JACQUARD'S LOOM :

- It was invented by Joseph Marie Jacquard in 1801.
- It was the first machine to use punched cards to control sequence of operations.
- It can create intricate patterns from a series of Punched cards.

6. DIFFERENCE ENGINE :

- In the early 1820s, it was designed by Charles Babbage who is known as "Father of Modern Computer".
- It was a mechanical computer which could perform simple calculations.
- It was a steam driven calculating machine designed to solve tables of numbers like logarithm tables.
- It could store the program and perform the results automatically.

7. ANALYTICAL ENGINE:

- This calculating machine was also developed by Charles Babbage in 1830.
- It was a mechanical computer that used punch-cards as input.
- It performs addition, subtraction, multiplication and division.
- It was capable of solving any mathematical problem and storing information as a permanent memory.

8. TABULATING MACHINE :

- Tabulating machines was invented in 1890, by Herman Hollerith, an American statistician.
- It was a mechanical tabulator based on punch cards.
- It could tabulate statistics and record or sort data or information.
- This machine was used in the 1890 U.S. Census.
- It was used in census. It gives a quick count of population.

9. DIFFERENTIAL ANALYZER :

- Differential Analyzer was the first electronic computer introduced in the United States in 1930.
- It was an analog device invented by Vannevar Bush.
- This machine has vacuum tubes to switch electrical signals to perform calculations.
- It could do 25 calculations in few minutes.

10. MARK I :

- In 1944, Mark I computer was built as a partnership between IBM and Howard.
- It was the first programmable digital computer.
- It was the first operating machine that could execute computations automatically.
- It is considered to be the real beginning of era of the modern computer.

Characteristics Of Computers :

- | | |
|---------------|-------------------------------|
| * Speed | * Memory and Storage Capacity |
| * Accuracy | * Programmable |
| * Reliability | * Easy Maintenance |
| * Diligence | * Resource Sharing |
| * Versatility | * Communication |

1) SPEED :

- A Computer process data at an extremely fast rate, at millions or billions of instructions per second.
- Computer can perform a huge task in a few seconds.
- It works with much higher speed and accuracy compared to humans.
- This speed depends upon the type of processor being used.
- The speed of computer is calculated in MHz(megahertz), that is one million instructions per second.
- At present, a powerful computer can perform billions of operations in a seconds.
- Modern computers can rapidly transmit and receive large amounts of data, enabling quick access to information and efficient communication.

2) ACCURACY :

- A computer system is capable of providing accurate results of computations.
- It can ensure consistently very high degree of accuracy in computations.
- Computers perform calculations and execute operations with a high level of accuracy.
- It process data according to the sequence of instructions. Hence, if input data and procedures are correct, the output will be consistently accurate.
- A computer can also generate errors only when the instructions given by the user are wrong.

3) RELIABILITY :

- Reliability is one of the significant characteristics of computer system, ensuring they perform their tasks consistently and dependably.
- Reliability is the measurement of the performance of the computer.
- The major reason behind the reliability of the computer is that, at hardware level, it doesnot require any human intervention between its processing operations.
- A computer system of the present generation is such reliable in terms of storage and processing.
- The results obtained by a computer are much accurate and faster than the results obtained manually.

4) DILIGENCE :

- A computer can repeatedly perform millions of large and complex calculations with constant speed and accuracy.
- It doesn't feel any fatigue, tiredness or lack of concentration.
- If four million calculations have performed, then the computer will perform the last four millionth calculation with same accuracy and same speed as the first calculation.
- It can work for hours without creating any errors.

5) VERSATILITY :

- A computer is said to be versatile in nature.
- It can perform multiple tasks simultaneously with equal ease.
- For Example : At one moment it can be used to draft a letter , another moment it can be used to play music and in between one can print a document as well.
- It is capable of handling multiple tasks simultaneously with same accuracy and efficiency.
- Computers are renowned for their remarkable versatility, capable of performing a wide range of tasks with ease.

6) MEMORY & STORAGE CAPACITY :

- Computers possess remarkable memory and storage capabilities, allowing them to handle vast amounts of data.
- Computer can store large amounts of data and can recall the information almost instantaneously.
- Storage capacity of present generation computers allow storage of large volume of data, including documents, music, animation clips, movies etc.
- The main memory of the computer is relatively small and it can hold only a certain amount of data ; therefore data are stored on secondary storage devices like magnetic tape or disks.
- Secondary storage devices like CDs, pen drives etc. are also used to store data.
- The storage capacity of modern computers can range from hundreds of gigabytes to several terabytes, enabling them to hold massive amounts of information.

7) PROGRAMMABLE :

- The computers can be programmed to work automatically.
- We can give them instructions in the form of code or programs to perform specific tasks.
- Programming languages like python, java or C allow us to write these instructions and communicate with computers.

8) EASY MAINTENANCE :

- Computer maintenance involves keeping a computer in a good state of repair and physical health through regular cleanings, hard drive updates and virus prevention.
- Computers can be easily managed.
- They can be easily repaired.
- The maintenance cost of computers is also less.

9) RESOURCE SHARING :

- In the initial stages of development , computers used to be isolated machines.
- With the tremendous growth in computer technologies, computers today have the capability to connect with each other.
- This has made the sharing of costly resources like printers possible.
- Apart from device sharing , data and information can also be shared among groups of computers, thus creating a large information and knowledge base.

10) COMMUNICATION :

- Most of computer have capability to communicate with other computer.
- Computer can be connected using communication device such as Modem.
- If it is connected together it is said to be a network.
- Internet is a good example of Network.

GENERATION OF COMPUTERS :

FIRST GENERATION (1940 – 1956) : VACUUM TUBES

- First – generation computers were vacuum tubes / thermionic valve – based machines.
- These computers used vacuum tubes for circuitry and magnetic drums for memory.
- A magnetic drum is a metal cylinder coated with magnetic iron oxide material on which data and programs can be stored.
- The input was based on punched cards and paper tape, and the output was in the form of printouts.
- First – generation computers relied on binary – coded language also called machine language (language of 0's and 1's) to perform operations and variable to solve only one problem at a time.
- Each machine was fed with different binary codes and hence, were difficult to program.
- This resulted in lack of versatility and speed.

Examples: ENIAC, EDVAC and UNIVAC.

Characteristics of First - generation Computers :

- These computers were based on vacuum tube technology.
- These were the fastest computing devices of their times (computation time was in milliseconds).
- These computers were very large and required a lot of space for installation.
- Since thousands of vacuum tubes were used, they generated a large amount of heat. Therefore, air conditioning was essential.
- These were non-portable and very slow equipments.
- They lacked in versatility and speed.
- They were very expensive to operate and used a large amount of electricity.
- These machines were unreliable and prone to frequent hardware failures. Hence, constant maintenance was required.
- Since machine language was used, these computers were difficult to program and use.
- Each individual component had to be assembled manually. Hence, commercial appeal of these computers was poor.

SECOND GENERATION (1956 – 1963) : TRANSISTORS

- Second-generation computers used transistors, which were superior to vacuum tubes.
- A transistor is made up of semiconductor material like germanium and silicon.
- Since a transistor is a small device, the physical size of computers was greatly reduced.
- Computers became smaller, faster, cheaper, energy efficient and more reliable than their predecessors.
- One of the major developments of this generation includes the progress from machine language to assembly language.
- Assembly language uses mnemonics (abbreviations) for instructions rather than numbers, for example, ADD for addition and MULT for multiplication.

Examples: PDP-8, IBM 1401 and IBM 7090.

Characteristics of Second - generation Computers :

- These machines were based on transistor technology.
- These were smaller as compared to the first-generation computers.
- The computational time of these computers was reduced to microseconds from milliseconds.
- These were more reliable and less prone to hardware failure. Hence, they required less frequent maintenance.
- These were more portable and generated less amount of heat.
- Assembly language was used to program computers. Hence, programming became more time-efficient and less cumbersome.
- Second-generation computers still required air conditioning.
- Manual assembly of individual components into a functioning unit was still required.

THIRD GENERATION (1964 TO Early 1970s) INTEGRATED CIRCUITS

- The development of the integrated circuit, also called an IC, was the trait of the third-generation computers..
- An IC consists of a single chip (usually silicon) with many components such as transistors and resistors fabricated on it. ICs replaced several individually wired transistors.
- This development made computers smaller in size, reliable and efficient.
- This allowed the device to run many different applications simultaneously with a central program that monitored the memory.

Examples: NCR 395 and B6500.

Characteristics of Third - generation Computers :

- These computers were based on IC technology.
- These were able to reduce the computational time from microseconds to Nanoseconds.
- These were easily portable and more reliable than the second- generation computers.
- These devices consumed less power and generated less heat. In some cases, air conditioning was still required.
- The size of these computers was smaller as compared to previous - generation computers.
- Since hardware rarely failed, the maintenance cost was quite low.
- Extensive use of high-level languages became possible.
- Commercial production became easier and cheaper.

FOURTH GENERATION (Early 1970s to 1985) : MICROPROCESSORS

- The fourth generation is an extension of third generation technology.
- A microprocessor is built on to a single piece of silicon, known as chip.
- The fourth-generation computers led to an era of Large Scale Integration (LSI) and VLSI technology.
- LSI technology allowed thousands of transistors to be constructed on one small slice of silicon material, whereas VLSI squeezed hundreds of thousands of components on to a single chip.
- Ultra Large Scale Integration (ULSI) increased that number to the millions.
- The fourth-generation computers became more powerful, compact, reliable and affordable.
- This generation also saw the development of the Graphical User Interfaces (GUIs), mouse and hand-held devices

Examples: Apple II, Altair 8800 and CRAY-1.

Characteristics of Fourth - generation Computers :

- These computers are microprocessor-based systems.
- These are very small in size.
- These are the cheapest among all the other generation computers.
- These are portable and quite reliable.
- These machines generate negligible amount of heat, hence do not require air conditioning.
- Hardware failure is negligible so minimum maintenance is required.
- The production cost is very low.
- The GUI and pointing devices enabled users to learn to use the computer quickly.
- Interconnection of computers led to better communication and resource sharing.

FIFTH GENERATION (1985 - Present) : ARTIFICIAL INTELLIGENCE

- The technology behind the fifth generation of computers is AI. It allows computers to behave like humans.
- It is often seen in programs like voice recognition, area of medicine, and entertainment.
- The speed is the highest, size is the smallest and area of use has remarkably increased within the fifth generation computers.
- Based on artificial intelligence, uses the Ultra Large-Scale Integration (ULSI) technology and parallel processing method.

(ULSI has millions of transistors on a single microchip and the Parallel processing method use two or more microprocessors to run tasks simultaneously).

1. Mega Chips:

- ✓ Fifth Generation computer use Super Large Scale Integrated (SLSI) chips, which will result in the production of microprocessor having millions of electronic components on a single chip.
- ✓ In order to store instructions and information, fifth generation computers require a great amount of storage capacity.
- ✓ Mega chips may enable the computer to approximate the memory capacity of the human mind.

2. Parallel Processing:

- ✓ Most computers today access and execute only one instruction at a time. This is called serial processing.
- ✓ However, a computer using parallel processing accesses several instructions at once and works on . then at the same time through use of multiple central processing units.

3. Artificial Intelligence (AI):

- ✓ It refer to a series of related technologies that tries to simulate and reproduce human behavior, including thinking, speaking and reasoning.
- ✓ AI comprises a group of related technologies: expert system (ES), natural language processing (NLP), speech recognition, vision recognition and robotics.

Characteristics of Fifth - generation Computers :

- ✓ The ULSI (Ultra Large Scale Integration) technology is used in this generation of computers.
- ✓ In this generation's computers, artificial intelligence has progressed.
- ✓ Parallel processing has advanced on these computers.
- ✓ Computers that are more portable and powerful.
- ✓ Computers are dependable and less expensive.
- ✓ It's easier to manufacture in a commercial setting.
- ✓ These computers are far quicker than previous generations.
- ✓ These computers are simpler to repair.
- ✓ These computers are substantially smaller in size than other generation computers.

Types Of Computers

Classification of the electronic computers may be based on either their principles of operation or their configuration, by configuration; we mean the size, speed of doing computation and storage capacity of a computer.

Classification based on Principles of Operation :

- 1)Analog
- 2)Digital
- 3)Hybrid

Classification of Computers based on Configuration :

1) Micro Computers :

- Desktop computers
- Portable computers
- Handheld Computer

2) Mini Computers

3) Mainframe Computers

4) Super Computers

1) ANALOG COMPUTERS :

- Analog computers work on the principle of measuring, in which the measurements obtained are translated into data rather than counting.
- Such computers do not deal directly with the numbers.
- It generally deals with physical variables such as voltage, pressure, temperature, speed etc.,
- It can directly accept the data from the measuring device without first converting it into numbers and codes.
- For example speedometer, mercury thermometer, etc.

2) DIGITAL COMPUTER

- Digital computers are those that operate with information, numerical or otherwise, represented in a digital form.
- Such computers process data into a digital value (in 0s and 1s). They give the results with accuracy and at a faster rate.
- All the data representation will be in the series of 0's and 1's.
- The digital computer is made for both general purpose and special purpose. Special purpose computer is one that is built for a specific application.
- General purpose computers are used for any type of applications.
- It can store different programs and do the jobs as per the instructions specified on those programs.

3) HYBRID COMPUTER

- A combination of computers those are capable of accepting inputs and outputs in both digital and analog signals.
- Now-a-days analog-to-digital and digital-to-analog converters are used for transforming the data into suitable form for either type of computation.
- Hybrid computers are mainly used for specialized tasks.
- For example, in hospital's automated intensive care unit, analog devices might measure the patient's temperature, blood pressure and other vital signs. These measurements which are in analog might then be converted into numbers and supplied to digital components in the system. These components are used to monitor the patient's vital sign and send signals if any abnormal readings are detected.
- A processor which is used in petrol pumps that converts the measurements of fuel flow into quantity and price is an example of a hybrid computer.

MICRO COMPUTER

- Micro computer is also called personal computer
- It is basically a general-purpose computer designed for individual use.
- It consists of a microprocessor as a central processing unit (CPU), memory, input unit, and output unit.
- These computers are widely used for general purpose such as word processing , web browsing , multimedia etc.
- It is used by one person at a time.
- For example, Laptops and desktop computers.
- The major types of personal computers are desktop computer and portable computer.

Characteristics :

- It is the smallest in size and easy to use.
- It is designed for personal use.

i) Desktop computers

- These computers can easily fit on a table or desktop, hence the name.

ii) Portable Computers

- Portable is a personal computer that can be carried from one place to other easily.
- Notebook computer and handheld computer (smart phone) are examples of portable computers.
- Notebook computer is also called laptop computer.
- Laptop computers are very small in size and can be placed easily on lap.

Handheld Computers (like smart phone) are also portable.

- Handheld computer is also known as palmtop computer.
- It easily fits in the hand of the user.

Uses of Micro Computer

- The PC is the most common type of computer used in the office.
- It is now widely used in many homes.
- These are also used for business and engineering application.

MINI COMPUTER

- Minicomputer is larger and more powerful computer than personal computer.
- It can execute five million instructions per second.
- It generally consists of two or more processors.
- Mini computer can serve up to 4000 connected users simultaneously.
- Minicomputers are used in places like institutes or departments for different work like billing, accounting, inventory management, etc. .
- Examples of mini computers are: VAX-8800, AS 400.

Characteristics :

- Its weight is low. Because of its low weight, it is easy to carry anywhere.
- less expensive than a mainframe computer.

Uses of Mini Computers :

- Mini computers are often used by small and medium-sized companies to provide centralized store of information.

MAIN FRAME COMPUTER

- Main Frame Computers are designed in such a way that they can support hundreds or thousands of users at the same time.
- It also supports multiple programs simultaneously. So, they can execute different processes simultaneously.
- All these features make the mainframe computer ideal for big organizations like banking, telecom sectors, etc., which process a high volume of data in general.
- It consists of multiple processors.
- Mainframe computers can serve up to 50,000 users at the same time.
- Examples of mainframe computers are NEC610, DEC 10.

Characteristics :

- A mainframe computer is a very large computer in size.
- It is more powerful than minicomputer.
- Massive Processing Power .
- Advanced Security Features.

Uses of Mainframe Computers :

- Mainframe computers are used in large organizations.
- For example, airlines use these computers for ticket reservation system.
- Used by large organizations such as banks , insurance companies and government institutions for large scale transaction processing etc.

SUPER COMPUTER

- Super computer is the most sophisticated, complex and advanced computer.
- They are the biggest and fastest computers (in terms of speed of processing data).
- Supercomputers are designed such that they can process a huge amount of data, like processing trillions of instructions or data just in a second.
- This is because of the thousands of interconnected processors in supercomputers.
- It is basically used in scientific and engineering applications such as weather forecasting and nuclear energy research.

Characteristics :

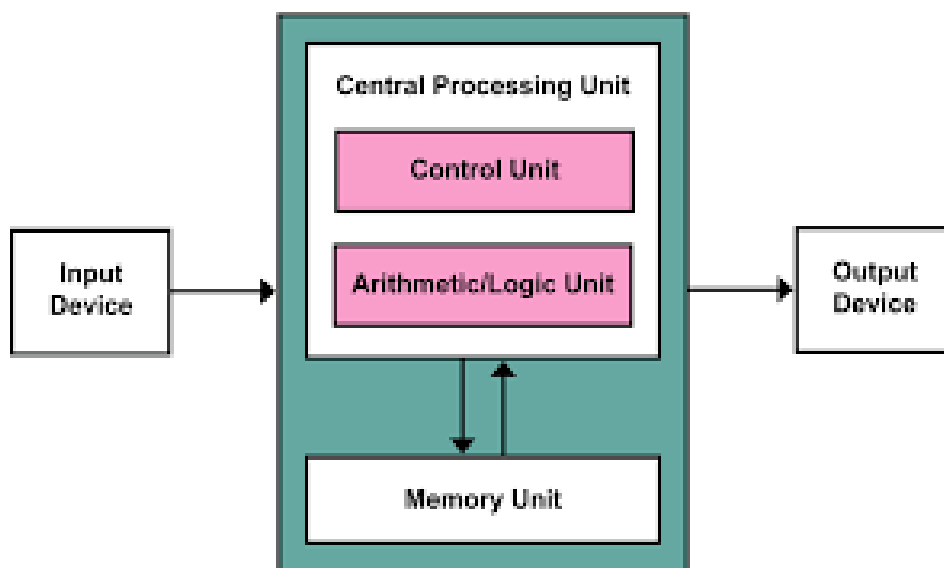
- Supercomputers are the computers that are the fastest and they are also very expensive.
- It can calculate up to ten trillion individual calculations per second, this is also the reason which makes it even faster.

Uses of Super Computers include:

1. Weather forecasting
2. Animated graphics like in Hollywood movies
3. Nuclear energy research
4. Space science etc.

VONN NEUMANN ARCHITECTURE

- Von Neumann architecture was first published by John von Neumann in 1945.
- His computer architecture design consists of a Control Unit, Arithmetic Logic Unit (ALU), Memory Unit, Registers and Inputs/Outputs.
- Von Neumann architecture is based on the stored-program computer concept, where instruction data and program data are stored in the same memory.



It is also known as ISA (Instruction Set Architecture) computer and is having three basic units:

1. The Central Processing Unit (CPU)
2. The Main Memory Unit
3. The Input / Output Device

1. Central Processing Unit: The central processing unit is defined as the it is an electric circuit used for the executing the instruction of computer program.

It has following major components:

- 1.1 Control Unit(CU)
- 1.2 Arithmetic and Logic Unit(ALU)
- 1.3 Variety of Registers

1.1 Control Unit

- This unit checks the correctness of the sequence of operations.
- It fetches the program instructions from the memory unit, interprets them and ensures correct execution of the program.
- It also controls the I/O devices and directs the overall functioning of the other units of the computer

1.2 Arithmetic and Logic Unit (ALU) :

- The arithmetic logic unit is that part of the CPU that handles all the calculations the CPU may need, e.g. Addition, Subtraction, Comparisons.
- It performs Logical Operations, Bit Shifting Operations, and Arithmetic operations.

1.3 Registers - Registers refer to high-speed storage areas in the CPU. The data processed by the CPU are fetched from the registers.

There are different types of registers used in architecture :-

- Accumulator: Stores the results of calculations made by ALU.
- Program Counter (PC): Keeps track of the memory location of the next instructions to be dealt with.
- Memory Address Register (MAR): It stores the memory locations of instructions that need to be fetched from memory or stored in memory.
- Memory Data Register (MDR): It stores instructions fetched from memory or any data that is to be transferred to, and stored in, memory.
- Current Instruction Register (CIR): It stores the most recently fetched instructions while it is waiting to be coded and executed.
- Instruction Buffer Register (IBR): The instruction that is not to be executed immediately is placed in the instruction buffer register IBR.

Buses – Data is transmitted from one part of a computer to another, connecting all major internal components to the CPU and memory, by the means of Buses. Types:

- Data Bus: It carries data among the memory unit, the I/O devices, and the processor.
- Address Bus: It carries the address of data (not the actual data) between memory and processor.
- Control Bus: It carries control commands from the CPU (and status signals from other devices) in order to control and coordinate all the activities within the computer.

Input / Output Devices –

- Program or data is read into main memory from the input device or secondary storage under the control of CPU input instruction.
- Output devices are used to output information from a computer.
- If some results are evaluated by the computer and it is stored in the computer, then with the help of output devices, we can present them to the user.

Von Neumann Bottleneck

- Whatever we do to enhance performance, we cannot get away from the fact that instructions can only be done one at a time and can only be carried out sequentially.
- Both of these factors hold back the competence of the CPU. This is commonly referred to as the ‘Von Neumann bottleneck’.
- We can provide a Von Neumann processor with more cache, more RAM, or faster components.

THE COMPUTER SYSTEM

- A computer can be viewed as a system, which consists of a number of interrelated components that work together with the aim of converting data into information.
- Every computer needs to be instructed exactly what to do and how to do. The instructions given to computers are called programs.
- Without programs, computers would be useless.
- The physical parts that make up a computer {the CPU, input, output and storage unit) are known as hardware.

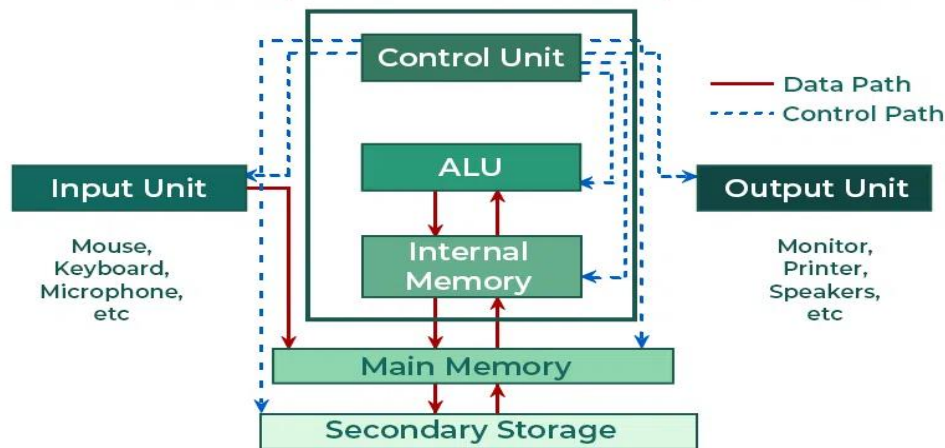
Components of a Computer System

There are several computer systems in the market with a wide variety of models and peripherals.

In general, a computer system comprises the following components:

- CPU: This unit performs processing of instructions and data inside the computer.
- Input Unit: This unit accepts instructions and data.
- Output Unit: This unit communicates the results to the user.
- Storage Unit: This unit stores temporary and final results.

Components of computer



❖ Central Processing Unit :

- CPU, also known as a processor, is the brain of the computer system that processes data (input) and converts it into meaningful information (output).
- It is referred to as the administrative section of the computer system that interprets the data and instructions, coordinates the operations, and supervises the instructions.
- The CPU works with data in discrete form, that is, either 1 or 0.
- It counts, lists, compares and rearranges the binary digits of data in accordance with the detailed program instructions stored within the memory.
- Eventually, the results of these operations are translated into characters, numbers and symbols that can be understood by the user.

The CPU itself has three parts:

1. Arithmetic Logic Unit (ALU):

- This unit performs the arithmetic (add, subtract) and logical operations (and, or).
- Whenever an arithmetic or logical operation is to be performed, the required data are transferred from the memory unit to the ALU, the operation is performed and the result is returned to the memory unit.
- The ALU performs the calculation only when needed and then sends it to the output devices.

2. Control Unit:

- This unit checks the correctness of the sequence of operations.
- It fetches the program instructions from the memory unit, interprets them and ensures correct execution of the program.
- It also controls the I/O devices and directs the overall functioning of the other units of the computer

3. Registers:

- These are the special-purpose, high-speed temporary memory units that can hold varied information such as data, instructions, addresses and intermediate results of calculations.
- Essentially, they hold the information that the CPU is currently working on.
- The registers can be considered as the CPU's working memory, an additional storage location that provides the advantage of speed.

❖ **Input, Output and Storage Unit :**

- An input unit accepts instructions and data from the user with the help of input devices such as keyboard, mouse, light pen, etc.
- Since the data and instructions entered through different input devices will be in different form, the input unit converts them into the form that the computer can understand.
- After this, the input unit supplies the converted instructions and data to the computer for further processing.
- The output unit performs just opposite to that of input unit.
- It accepts the output (which is in machine-coded form) produced by the computer, converts them into the user-understandable form and supplies the converted results to the user with the help of an output device such as printer, monitor and plotter.
- Besides, a computer system consist a storage unit to store the input entered through the input unit before processing starts and to store the results produced by the computer before supplying them to the output unit.
- The storage unit of a computer comprises two types of memory/storage: primary and secondary
- The primary memory, also called the main memory, is the part of a computer that holds the instructions and data currently being processed by the CPU, the intermediate results produced during the course of calculations and the recently processed data.
- However, the primary memory is quite expensive and has a limited storage capacity.
- Due to the limited size of the primary memory, a computer employs the secondary memory, which is extensively used for storing data and instructions.
- It supplies the stored information to the other units of the computer as and when required.
- It is less expensive and has higher storage capacity than the primary memory.

INPUT DEVICES : Following are few of the important input devices which are used in a computer:

- Keyboard
- Mouse
- Joy Stick
- Light pen
- Track Ball
- Scanner
- Graphic Tablet
- Microphone
- Magnetic Ink Card Reader MICR
- Optical Character Reader OCR
- Bar Code Reader
- Optical Mark Reader OMR

Keyboard :

- Keyboard is the most common and very popular input device which helps in inputting data to the computer.
- The layout of the keyboard is like that of traditional typewriter, although there are some additional keys provided for performing additional functions.
- Keyboards are of two sizes 84 keys or 101/102 keys, but now keyboards with 104 keys or 108 keys are also available for Windows and Internet.

The keys on the keyboard are as follows:

- **Typing Keys:** These keys include the letter keys A - Z and digit keys 0 - 9 which generally give same layout as that of typewriters.
- **Numeric Keypad:** It is used to enter numeric data or cursor movement. Generally, it consists of a set of 17 keys that are laid out in the same configuration used by most adding machines and calculators.
- **Function Keys:** The twelve function keys are present on the keyboard which is arranged in a row at the top of the keyboard. Each function key has unique meaning and is used for some specific purpose.
- **Control keys:** These keys provide cursor and screen control. It includes four directional arrow keys. Control keys also include Home, End, Insert, Delete, Page Up, Page Down, Control Ctrl, Alternate Alt and Escape Esc.
- **Special Purpose Keys:** Keyboard also contains some special purpose keys such as Enter, Shift, Caps Lock, Num Lock, Space bar, Tab, and Print Screen.

Mouse :

- Mouse is most popular pointing device.
- It is a very famous cursor - control device having a small palm size box with a round ball at its base which senses the movement of mouse and sends corresponding signals to CPU when the mouse buttons are pressed.
- Generally it has two buttons called left and right button and a wheel is present between the buttons.
- Mouse can be used to control the position of cursor on screen, but it cannot be used to enter text into the computer.

Advantages

- Easy to use
- Not very expensive
- Moves the cursor faster than the arrow keys of keyboard.

Joystick :

- Joystick is also a pointing device which is used to move cursor position on a monitor screen.
- It is a stick having a spherical ball at its both lower and upper ends.
- The joystick can be moved in all four directions. The function of joystick is similar to that of a mouse.
- It is mainly used in Computer Aided Designing CAD and playing computer games.

Light Pen :

- Light pen is a pointing device which is similar to a pen.
- It is used to select a displayed menu item or draw pictures on the monitor screen.
- When the tip of a light pen is moved over the monitor screen and pen button is pressed, its photocell sensing element detects the screen location and sends the corresponding signal to the CPU.

Track Ball :

- Track ball is an input device that is mostly used in notebook or laptop computer, instead of a mouse.
- This is a ball which is half inserted and by moving fingers on ball, pointer can be moved.
- Since the whole device is not moved, a track ball requires less space than a mouse.
- A track ball comes in various shapes like a ball, a button and a square.

Scanner :

- Scanner is an input device which works more like a photocopy machine.
- It is used when some information is available on a paper and it is to be transferred to the hard disc of the computer for further manipulation.
- Scanner captures images from the source which are then converted into the digital form that can be stored on the disc.
- These images can be edited before they are printed.

Digitizer :

- Digitizer is an input device which converts analog information into digital form.
- Digitizer can convert a signal from the television or camera into a series of numbers that could be stored in a computer.
- Digitizer is also known as Tablet or Graphics Tablet because it converts graphics and pictorial data into binary inputs.
- A graphic tablet as digitizer is used for doing fine works of drawing and image manipulation applications.

Microphone :

- Microphone is an input device to input sound that is then stored in digital form.
- The microphone is used for various applications like adding sound to a multimedia presentation or for mixing music.

Magnetic Ink Card Reader (MICR) :

- MICR input device is generally used in banks because of a large number of cheques to be processed every day.
- The bank's code number and cheque number are printed on the cheques with a special type of ink that contains particles of magnetic material that are machine readable.
- This reading process is called Magnetic Ink Character Recognition MICR.
- The main advantages of MICR are that it is fast and less error prone.

Optical Character Reader (OCR) :

- OCR is an input device used to read a printed text.
- OCR scans text optically character by character, converts them into a machine-readable code and stores the text on the system memory.

Bar Code Readers :

- Bar Code Reader is a device used for reading bar coded data which are in the form of light and dark lines.
- Bar coded data is generally used in labelling goods, numbering the books etc.
- It may be a hand held scanner or may be embedded in a stationary scanner.
- Bar Code Reader scans a bar code image, converts it into an alphanumeric value which is then fed to the computer to which bar code reader is connected.

Optical Mark Reader (OMR) :

- OMR is a special type of optical scanner used to recognize the type of mark made by pen or pencil.
- It is used where one out of a few alternatives is to be selected and marked.
- It is specially used for checking the answer sheets of examinations having multiple choice questions.

OUTPUT DEVICES :

Following are few of the important output devices which are used in a computer.

- ✓ Monitors
- ✓ Printers
- ✓ Headphones
- ✓ Speakers
- ✓ Projector
- ✓ Plotter

Monitors

- Monitors, commonly called as Visual Display Unit VDU, are the main output device of a computer.
- It forms images from tiny dots, called pixels that are arranged in a rectangular form.
- The sharpness of the image depends upon the number of pixels.
- There are two kinds of viewing screen used for monitors :
 - a) Cathode Ray Tube (CRT)
 - b) Flat Panel Display

a) Cathode - Ray Tube (CRT) Monitor

- The CRT display is made up of small picture elements called pixels. The smaller the pixels, the better the image clarity, or resolution.
- A finite number of characters can be displayed on a screen at once.
- Most screens are capable of displaying 80 characters of data horizontally and 25 lines vertically.

There are some disadvantages of CRT:

Large in Size

High power consumption

b) Flat - Panel Display Monitor

- The flat-panel display refers to a class of video - devices that have reduced volume, weight and power requirement in comparison to the CRT.
- Current uses of flat-panel displays include calculators, video games, monitors, laptop computer, graphics display.

The flat-panel display is divided into two categories:

Emissive Displays - The emissive displays are devices that convert electrical energy into light. Example are plasma panel and LED (Light Emitting Diodes).

Non-Emissive Displays - The Non-emissive displays use optical effects to convert sunlight or light from some other source into graphics patterns, Example is LCD (Liquid - Crystal Device).

Printers

- Printer is an output device, which is used to print information on paper.
- There are two types of printers:

Impact Printers

Non-Impact Printers

1) Impact Printers

- The impact printers print the characters by striking them on the ribbon which is then pressed on the paper.

Characteristics of Impact Printers :

- Very low consumable costs
- Very noisy
- Useful for bulk printing due to low cost.

These printers are of two types

I. Character printers

II. Line printers

I. Character Printers

- Character printers are the printers which print one character at a time.
- These are further divided into two types:

i) Dot Matrix Printer (DMP)

ii) Daisy Wheel

i) Dot Matrix Printer

- In the market one of the most popular printers is Dot Matrix Printer.
- These printers are popular because of their ease of printing and economical price.
- Each character printed is in form of pattern of dots and head consists of a Matrix of Pins of size $5 * 7$, $7 * 9$, $9 * 7$ or $9 * 9$ which come out to form a character that is why it is called Dot Matrix Printer.

Advantages

- Inexpensive
- Widely Used
- Other language characters can be printed

Disadvantages

- Slow Speed
- Poor Quality

ii) Daisy Wheel

- Head is lying on a wheel and pins corresponding to characters are like petals of Daisy flower that is why it is called Daisy Wheel Printer.
- These printers are generally used for word-processing in offices which require a few letters to be sent here and there with very nice quality.

Advantages

- More reliable than DMP
- Better quality
- The fonts of character can be easily changed

Disadvantages

- Slower than DMP Noisy
- More expensive than DMP.

II. Line Printers

- Line printers are the printers which print one line at a time.
- These are of further two types

a) Drum Printer b) Chain Printer

a) Drum Printer :

- This printer is like a drum in shape so it is called drum printer.
- Type of line printer that uses rotating drum to print text onto paper.
- In this printer , all the characters are not printed at one time but their printing speed is very high.
- One rotation of drum prints one line.
- Drum printers are fast in speed and can print 300 to 2000 lines per minute.

Advantages

- Very high speed

Disadvantages

- Very expensive
- Characters fonts cannot be changed.

b) Chain Printer :

- In this printer, chain of character sets is used so it is called Chain Printer.
- Chain printer are one of the fastest impact printers that can produce up to 400 to 2500 characters per second.

Advantages

- Character fonts can easily be changed.
- Different languages can be used with the same printer.

Disadvantages

- Noisy

2) Non - impact Printers :

- Non-impact printers print the characters without using ribbon.
- These printers print a complete page at a time so they are also called as Page Printers.
- These printers are of two types
 - i) Laser Printers
 - ii) Inkjet Printers

Characteristics of Non-impact Printers :

- Faster than the impact printers.
- They are not noisy.
- High quality
- Support many fonts and different character size.

i) Laser Printers

- These are non-impact page printers.
- They use laser lights to produce the dots needed to form the characters to be printed on a page.

Advantages

- Very high speed
- Very high quality output
- Give good graphics quality
- Support many fonts and different character size.

Disadvantages

- Expensive.
- Cannot be used to produce multiple copies of a document in a single printing.

ii) Inkjet Printers

- Inkjet printers are non-impact character printers based on a relatively new technology.
- They print characters by spraying small drops of ink onto paper.
- Inkjet printers produce high quality output with presentable features.
- Colour printing is also possible.
- Some models of Inkjet printers can produce multiple copies of printing also.

Advantages

- High quality printing.
- More reliable.

Disadvantages

- Expensive as cost per page is high.
- Slow as compared to laser printer.

Plotter Printers

- Plotter is an output device that draws pictures on paper based on commands from a computer.
- Plotters differ from printers in that they draw lines using a pen. As a result, they can produce continuous lines, whereas printers can only simulate lines by printing a closely spaced series of dots.
- Multicolor plotters use different-colored pens to draw different colors.
- In general, plotters are considerably more expensive than printers.
- They are used in engineering applications where precision is mandatory.

Speakers :

- The speakers are the output units.
- The sound signals from analog/digital are converted into audible frequency in the speakers and produce voice output (audio data).
- Using speakers along with speech synthesizer software, the computer can provide voice output.
- Voice output has become very common in many places like airlines, banks, automatic telephone enquiry system etc.
- Users can also hear music/ songs using the voice output system.
- The latest speakers are crystal clear audio. Walkman speakers are smaller in size.

Storage devices :

- These are essential hardware components that store digital data, allowing users to save, retrieve, and manage information.
- They vary in speed, capacity, and permanence, catering to different storage needs.

Types of Storage Devices:

1. Hard Disk Drive (HDD) :

- A traditional storage device that uses magnetic disks to read and write data.

Types:

- ❖ Internal HDD : Installed inside the computer, typically used for primary storage.
- ❖ External HDD : Portable drives that connect via USB or other interfaces for additional storage.

Function: Commonly used for operating systems, applications, and large data files due to their high capacity and cost-effectiveness.

2) Solid State Drive (SSD)

- A storage device that uses flash memory to store data, offering faster access times than HDDs.

Function: Used in laptops and desktops and overall system performance.

3) Optical Disc Description:

- A storage medium that uses laser technology to read and write data.

Types:

- ❖ CD (Compact Disc): Stores up to 700 MB of data, used for music and software distribution.
- ❖ DVD (Digital Versatile Disc): Stores 4.7 GB to 8.5 GB of data, commonly used for movies and software.

Function: Used for media playback, data backup, and software installation.

4) USB Flash Drive Description :

- A portable storage device that connects via USB, using flash memory to store data.

Types:

- ❖ Standard USB Flash Drive: Varies in capacity, typically ranging from 4 GB to 1 TB.
- ❖ OTG (On The Go) Flash Drive: Can connect to mobile devices directly.

Function: Convenient for data transfer, backup, and portable storage.

5) Memory Card :

- A small storage devices used in cameras , smartphones and other portable devices.

Types :

- ❖ SD Card :Commonly used in cameras and smartphones , with capacities ranging from 2 GB to 1 TB.
- ❖ Micro SD Card : smaller version of SD Cards , widely used in mobile devices.

Function : Provides expandable storage for devices and easy transfer between devices.

6) External SSD :

- A portable SSD that connects to devices via USB.

Function : Offers fast , portable storage for large files , ideal for photographers , videographers.

7) Network Attached Storage (NAS) :

- A dedicated file storage device connected to a network , allowing multiple users to access data.

Function : Used for centralized storage , backup solution.

8) Tape Drives :

- Magnetic Storage devices that uses tape for data storage, primarily used for backup and archiving.

Function : Suitable for long term data due to its high capacity

PROCESSOR & MAIN MEMORY:

- The Central Processing Unit (CPU) often referred to as the "brain" of the computer, is the primary component responsible for executing instructions and processing data in a computer system.

Functions of the CPU:

- 1) **Instruction Execution :** The CPU executes instructions from computer programs, performing basic arithmetic, logic, control, and input/output operations. It interprets instructions fetched from memory and translates them into actions.
- 2) **Data Processing:** The CPU processes data by performing calculations and comparisons, manipulating Data to produce results. This includes operations like addition, subtraction, multiplication and division, as well as logical operations (AND, OR, NOT).
- 3) **Control Unit:** The CPU includes a control unit that coordinates the activities of the other components of the computer. It fetches instructions from memory, decodes them, and sends control signals to execute them, ensuring the proper sequence of operations.
- 4) **Data Storage and Retrieval :** The CPU temporarily stores data and instructions in its registers, which are small, high-speed storage locations. It retrieves data from the main memory (RAM) and writes results back to memory when needed.

Components of the CPU :

1. Arithmetic Logic Unit (ALU) :

- The ALU is responsible for performing all arithmetic (addition, subtraction) and logical (comparison operations) calculations.
- It takes inputs from the registers and outputs results back to the registers.

2. Control Unit (CU) :

- The control unit manages the flow of data within the CPU and between the CPU and other components of the computer.
- It interprets instructions and generates control signals to execute them.

3. Registers :

- Registers are small, fast storage locations within the CPU that hold data and instructions temporarily during processing.
- They include general-purpose registers, instruction registers, and accumulator registers, facilitating quick access to data.

4. Cache Memory :

- Cache memory is a small amount of high-speed memory located within the CPU or close to it.
- It stores frequently accessed data and instructions, allowing for quicker retrieval and reducing the time needed to access the slower main memory.

5. System Bus Interface :

- The CPU communicates with other components of the computer via a system bus, which includes the data bus, address bus, and control bus.
- This interface allows the CPU to send and receive data and instructions from memory and input/output devices.

Architecture of the Processor :

- The architecture of the processor (CPU) defines its design and functionality. It consists of several key components and organization strategies:

1. Components of Processor Architecture:

- **Arithmetic Logic Unit (ALU) :** The ALU performs all arithmetic and logical operations. It takes input data from the registers, processes it, and sends the result back to the registers.
- **Control Unit (CU) :** The CU carries the operations of the CPU. It interprets the instructions from memory, directs the operation of the ALU, and coordinates the movement of data between the CPU, memory, and I/O devices.
- **Registers :** Registers are small, high-speed storage locations within the CPU that hold data, instructions, and addresses temporarily. Common registers include the Accumulator (ACC), Program Counter (PC), Instruction Register (IR), and various general-purpose registers.
- **Cache Memory :** Cache is a small-sized type of volatile memory that provides high-speed access to frequently used data and instructions.

2) Processor Organization :

- **Data Path** : The data path refers to the internal pathways that connect various components of the processor, enabling data transfer and processing. It includes buses for data movement between registers, ALU, and memory.
- **Control Path** : The control path manages the control signals required for the operation of various components within the CPU. It ensures that the correct operations are executed at the right time.
- **Instruction Pipeline** : Many modern processors use pipelining to improve instruction throughput. It divides instruction execution into stages (fetch, decode, execute, memory access, write back), allowing multiple instructions to be processed simultaneously at different stages.

Architecture of Main Memory

- Main memory, also known as RAM (Random Access Memory), is critical for the CPU's operation, as it stores data and instructions that the CPU needs while performing tasks.

1) Components of Main Memory Architecture :

- **Memory Cells** : The basic unit of memory, each cell can store a single bit of data (0 or 1).
- **Memory Addressing** : Each memory cell has a unique address, allowing the CPU to retrieve or store data at specific locations.
- **Memory Controller** : This component manages the flow of data between the CPU and memory. It translates addresses from the CPU into control signals that access the appropriate memory locations.

2) Memory Organization :

- **Word Size** : Memory is organized into words, which are groups of bits (typically 8, 16, 32, or 64 bits) that the CPU can process at once. The word size affects the amount of data that can be retrieved or stored in a single operation.
- **Dynamic and Static RAM** : Main memory typically consists of DRAM (Dynamic RAM), which requires periodic refreshing to maintain data, and SRAM (Static RAM), which is faster and more expensive but does not require refreshing.

COMPUTER MEMORY

- A memory is just like a human brain.
- It is used to store data and instructions.
- Computer memory is the storage space in computer where data is to be processed and instructions required for processing are stored.
- The memory is divided into large number of small parts called cells.
- Each location or cell has a unique address which varies from zero to memory size minus one.
- There are various units which are used to measure computer memory

Bit - Smallest unit of computer memory

Byte - 8 bit = 1 byte

Kilobyte - 1024 byte = 1 kb

Megabyte - 1024 kb = 1 mb

Gigabyte - 1024 mb = 1 gb

Terabyte - 1024 gb = 1 tb

TYPES OF COMPUTER MEMORY

- Memory is the best essential element of a computer because computer can't perform a simple task.
- The performance of the computer mainly based on memory and CPU.
- Memory is internal storage media of computer that has several names such as majorly categorized into two types .
- They are : Primary Memory
Secondary Memory

Primary Memory

- Primary Memory is a computer memory that a processor or computer accesses first or directly.
- Primary memory temporarily stores data that the CPU is executing, including instructions and result of those instructions.
- Primary memory as also known as primary storage or main memory.
- Types of primary memory:

RAM (Random Access Memory)

ROM (Read Only Memory)

Random Access Memory (RAM):

- It performs both read and writes operations on memory.
- If power failures happened in systems during memory access then you will lose your data permanently.
- So, RAM is volatile memory.

Characteristics of RAM :

- It allows writing, retrieving, erasing and writing again of data on it that can be accessed by user.
- It is a volatile memory, it loses its contents once the computer is shut down or goes off due to power problems. i.e. work as you save your work.

TYPES OF RAM :

1) SRAM (Static Random Access Memory) :

- Static random-access memory (SRAM) is a form of a semiconductor.
- Uses flip flop to store the data which makes it faster and more reliable than DRAM
- SRAM exhibits data remainance, but it is still volatile in the conventional sense that data is eventually lost when the memory is not powered.

2) DRAM (Dynamic Random Access Memory) :

- Stores each bit of data in a capacitor , which needs to be refreshed periodically , to retain information.
- It is slower than SRAM but allows for higher densities and is more cost effective.

Read Only Memory (ROM) :

- ROM is permanent memory location that offers huge types of standards to save data.
- But it work with read only operation.
- No data lose Happen whenever power failure occurs during the ROM memory work in computers.

Characteristics of ROM :

- Data is not lost when power is lost (non volatile).
- Once data is stored, one can only retrieve it but cannot edit it except if it was written on a special type of ROM.

Types of ROM :

1) PROM (Programmable Read Only Memory) :

- Maintains large storage media but can't offer the erase features in ROM.
- It maintains PROM chips to write data once and read many.
- The programs or instructions designed in PROM can't be erased by other programs.

2) EPROM (Erasable Programmable Read Only Memory) :

- designed for recover the problems of PROM and ROM.
- Users can delete the data of EPROM thorough pass on ultraviolet light.

3) EEPROM (Electrically Erasable Programmable Read Only Memory) :

- Similar to the EPROM but it uses electrical beam for erase the data of ROM.

Secondary Memory

- Secondary memory is external and permanent memory that is useful to store the external storage media such as floppy disk, magnetic disks, magnetic tapes and etc cache devices.

The Characteristics of Secondary Memory

- ✓ Large capacity
- ✓ Portability
- ✓ Durability
- ✓ Cost
- ✓ Non-volatility

Types of Secondary Memory:

- Magnetic Storage Media
- Optical Storage Media
- Flash Memory

Magnetic Storage Media :

- Magnetic Storage Media and devices store data in the form of tiny magnetized dots.
- These dots are created read and erased using Magnetic fields created by very tiny electromagnets.

Types Of Magnetic Storage Media :

- 1) Magnetic Tape
- 2) Hard Disk
- 3) Floppy Disk

1) Magnetic Tape :

- Magnetic tape drives have been in use longer than Floppies.
- Primarily used for back up storage .
- Tape drives are now capable of backing upto 3 terabytes of information

2) Hard Disk :

- Most Computer hard drives are magnetic .
- The hard drive is where the computer operating system and programs are stored .
- Storage capacity of hard disk varies by the drive size , but a typical hard drive for a new computer have storage capacity of 500 GB to 1 TB.

3) Floppy Disk :

- Floppy disk is a magnetic storage medium for computer systems.
- The floppy disk is composed of a thin, flexible magnetic disk sealed in a square plastic carrier.
- A floppy disk is also referred to simply as a floppy.
- Since the early days of personal computing, floppy disks were widely used to distribute software, transfer files, and create back-up copies of data.

Types of floppy disk :

8 Inch Floppy disk

5.25.Inch Floppy disk

3.5 Inch Floppy disk

Optical Storage Media :

- Optical Storage is any storage method in which data is written and read with a laser for archival or back up purposes.

Example : CDs , DVDs

Cache memory :

- It also called as CPU memory which is RAM that computer microprocessor can access more quickly than it can access regular RAM.
- It stores frequently used data and instruction, so that CPU can access them quickly.
- This Memory is typically integrated directly with the CPU chip or placed on a separate Chip that has a Separate BUS interconnect with the CPU.

Flash Memory :

- Flash memory is secondary memory and so it is not volatile which means it persists the data even if there is not an electrical supply provided.
- ROM operation can only one time write and many times read and we can't erase it. But Flash Memory can be erased multiple times and update the data or program integrated into it.
- Flash memory devices can store large amounts of data, from a few GB (Gigabytes) to several TB(Terabytes).

HARDWARE & SOFTWARE

HARDWARE

- Hardware refers to the physical elements of a computer.
- This is also sometime called the machinery or the equipment of the computer.
- Examples of hardware in a computer are the keyboard, the monitor, the mouse and the central processing unit.
- A computer's hardware is comprised of many different parts, but perhaps the most important of these is the motherboard.
- The motherboard is made up of even more parts and it control the computer.
- Hardware and software are interconnected, without software, the hardware of a computer would have no function.
- However, without the creation of hardware to perform tasks directed by software via the central processing unit, software would be useless.

SOFTWARE

- Software, commonly known as programs or apps, consists of all the instructions that tell the hardware how to perform a task.
- These instructions come from a software developer in the form that will be accepted by the platform (operating system + CPU).
- For example, a program that is designed for the Windows operating system will only work for that specific operating system.
- Software is capable of performing many tasks, as opposed to hardware which can only perform mechanical tasks that they are designed for.

Practical computer systems divide software systems into two major classes : System Software

Application Software

I. System software :

- Type of computer program designed to run a computer's hardware and application programs.
- System software is almost always pre-installed on your computer.
- There are two types in system software, they are
 - 1) Operating system
 - 2) Utility software

1) Operating System :

- The operating system is part of the system software.
- All computers have an operating system, they cannot function without one.
- The operating system is a program that allows applications software to communicate with the hardware.
- Examples of operating systems are Windows Vista, Windows XP, Unix, Linux and MacOS.

An operating system has many tasks. Here are just a few of them:

- Sorting out where to store data on disk drives
- Dealing with security - user names and passwords
- Organizing files and folders
- Managing data transfer from the CPU to the peripherals e.g. printer, monitor
- Deals with saving, deleting, opening, closing files

2) Utility Software :

- A Utility program is designed to do one or perhaps two tasks very well and nothing else.
- For example, you may want to compress a file to let you save it on to a flash drive.
- For this task you would choose to use a file compression utility program.
- For example Windows has a built in 'Zip' compression utility you can use to compress a file or folder.

Here are some very common tasks that are carried out by various Utility programs

- File sorting
- File renaming
- File conversion (e.g. convert a sound file to MP3)
- File repair
- Backing up data

II. Application software:

- Allows users to accomplish one or more tasks.
- It includes word processing, web browsing and almost any other task for which you might install software. (Some application software is pre-installed on most computer systems.)
- Software is generally created (written) in a high-level programming language, one that is readable by people.
- These high-level instructions are converted into "machine language" instructions, represented in binary code, before the hardware can "run the code".
- When you install software, it is generally already in this machine language, binary form.

- There are two types in application software, they are
 - i) General Purpose Software
 - ii) Specific Software

i) General Purpose Software :

- A general purpose application is the sort of software that you use at home and school.
- Examples include word processors, spread sheets, databases, graphics packages etc.
- This type of software provides many features that the majority of users will want e.g. formatting text, creating charts, organizing tables.

There are several good reasons for using general purpose software:

- It is relatively cheap.
- It is easily available from most computer shops.
- It will have been thoroughly tested so there won't be any serious problems or bugs.
- There will be lots of user support i.e. books, user guides, online help and discussion forums on the Internet.

ii) Specific Software :

- Special purpose application software is a type of software created to execute one specific task.
- For example a camera application on your phone will only allow you to take and share pictures.
- Another example would be a chess game, it would only allow you to play chess.

TRANSLATORS :

- In computing, a translator is a type of program that converts code written in one programming language into another language.
- This is essential because the code written by a programmer (usually in a high-level language) needs to be translated into machine code so that a computer's hardware can execute the instructions.

Compiler :

- A compiler is a special program that processes statements written in a particular programming language and turns them into machine language or "code" that a computer's processor uses.
- Typically, a programmer writes language statements in a language such as Pascal or C one line at a time using an editor.

Interpreter :

- The software by which the conversion of the high-level instructions is performed line-by-line to machine-level language, other than compiler and assembler, is known as INTERPRETER.
- The interpreter in the compiler checks the source code line-by-line and if an error is found on any line, it stops the execution until the error is resolved.
- Error correction is quite easy for the interpreter as it provides a line-by-line error. But the program takes more time to complete the execution successfully.

Assembler :

- An assembler is a type of computer program that interprets software programs written in assembly language into machine language, code and instructions that can be executed by a computer.
- An assembler is sometimes referred to as the compiler of assembly language.
- It also provides the services of an interpreter.

Linker :

- A linker is responsible for combining various object files (generated by the compiler or assembler) into a single executable file.
- Object files may come from different source code files.
- The linker is used during the final stages of compiling large programs, especially those using multiple.

Loader :

- A loader is a system software component that loads an executable file into memory and prepares it for execution by the CPU.
- Example: When you run a program on your operating system, the loader brings the program into memory and starts it.
- The loader is called when the user or system initiates the execution of a compiled program.

Editor :

An editor may refer to any of the following:

- In general, an editor refers to any program capable of editing files.
- Good examples are image editors, such as Adobe Photoshop, and sound editors, such as Audacity.
- The term editor is commonly used to refer to a text editor, which is a software program that allows users to create or manipulate plain text computer files.
- They are often used in the field of computer programming.

COMPUTER LANGUAGES :

Machine Level language :

- Machine language is the lowest-level programming language.
- Machine languages are the only languages understood by computers.
- It consisting of binary code (Os and 1s) that the computer's CPU can execute directly.
- Each machine has its own specific machine language based on its architecture.

Merits :

- Fast execution : Programs written in machine language are executed directly by the hardware without any translation, making them the fastest to execute.
- Full control over hardware : Provides direct access to the processor and memory, allowing optimization at the lowest level.

Demerits :

- Difficult to understand and write : Writing in binary is error-prone and difficult to debug.
- Non-portable : Machine language is architecture-specific, meaning code written for one machine will not work on another.

Example : A binary instruction like 11001100 might tell the processor to move a value from one register to another, but it's incomprehensible to most humans.

Assembly Level Language :

- Assembly language provides a symbolic representation of machine code using mnemonics (human-readable codes) and is specific to a particular computer architecture.
- It is not just a single language, but rather a group of languages.
- Each assembly instruction corresponds directly to a machine-level instruction.

Merits :

- Efficient : Assembly programs can be optimized to run faster and use fewer resources, making them suitable for systems programming or hardware-level tasks.
- Easier to modify and debug than machine code : The use of mnemonics makes it slightly more understandable.

Demerits :

- Low-level and complex : Though easier than machine language, assembly is still hard to write, understand, and maintain compared to high-level languages.
- Time-consuming : Writing large programs in assembly takes a lot of time, as every single instruction must be manually coded.

Example : An assembly language instruction such as `MOV AX, 5` moves the value 5 into the register AX.

High Level (Programming) Language :

- High-level languages (HLLs) are programming languages that are much closer to human languages and are independent of the architecture of the machine.
- They allow the programmer to write instructions using abstract concepts that are automatically translated into machine code by a compiler or interpreter.

Example: Java, C, Python, C++, C#, Java script

Example: A high-level language like Python allows a programmer to write `print("Hello, World!")`, which will print a message to the screen.

Merits :

- Easy to learn and use : High-level languages resemble human languages or logical constructs, making them much easier to learn and use.
- Portable: High-level languages are not tied to any specific architecture, allowing the same code to be run on different machines with minimal or no changes.
- Faster development : The ability to express complex tasks in fewer lines of code makes program development much faster.
- Better debugging and maintenance: High-level languages often come with tools that assist in debugging, error handling, and maintaining code.

Demerits :

- Less efficient : Programs written in high-level languages generally do not run as fast or use as little memory as those written in assembly or machine languages .
- Dependency on translators : High-level programs need to be translated (compiled or interpreted) into machine code before they can be executed, which can introduce inefficiencies.

PLANNING A COMPUTER PROGRAM:

- Refers to the process of designing and outlining the structure, functionality, and flow of a software application before actual coding begins.
- Algorithm, flowchart, and pseudocode are essential tools used in the planning and design phase computer programming.
- Each gives a specific purpose and can help in effectively outlining a program's before coding begins.

1. ALGORITHM :

- An algorithm is a step-by-step procedure or set of rules designed to solve a specific problem perform a task.

OR

An algorithm is a defined sequence of instructions that outlines a process to solve a problem systematically.

Characteristics :

- Clear and Absolute : Each of its steps should be clear in all aspects and must lead to only one meaning.
- Well-defined inputs and outputs : An algorithm should specify the inputs it requires and the outputs it produces.
- Finite steps : It must terminate after a certain number of steps, submitting a result.
- Effectiveness : Each step should be simple enough to be executed.

Example: An algorithm to find the sum of two numbers

1. Start
2. Read a and b
3. $\text{Sum} = a + b$
4. Print Sum
5. Stop

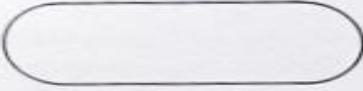
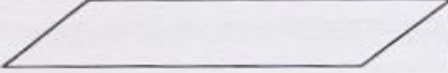
2. FLOWCHART

- A flowchart is a visual representation of the sequence of steps and decisions needed to solve a problem. It uses standardized symbols to denote different types of actions or steps.

OR

Flowchart is a graphical representation of an algorithm.

- Normally algorithm is a first step in program development, where as flowchart is designed once algorithms are completed. Algorithm is step by step written procedure of a program where as flow chart is flow of the same algorithm in picture or symbol form.

SYMBOLS USED IN FLOWCHART		
SYMBOL	NAME OF SYMBOL	FUNCTION
	Oval/ Start/End Terminal/Terminator	Use to start and stop the program. It will be used only twice in the program.
	Parallelogram/ Data/Input or Output	Used to input and output the data
	Decision/ Diamond	It makes decision and gives two path, if true one path and if false another path
	Flow lines/ Arrows	Shows flow of the chart
	Circle/ On-page connector/reference	On-page connector/reference

ADVANTAGES AND DISADVANTAGES OF FLOWCHART:

Advantages of flowchart

- Easy to make
- Mistakes can be easily identified
- Communication becomes effective and easy to understand
- Analysis becomes effective.

Disadvantages of flowchart

- Difficulty in presenting complex programs and tasks.
- It's a time-consuming process
- Difficult to understand for people who don't know flowchart symbols
- Complex when program size grows

ALGORITHM AND FLOW CHART TO FIND LARGEST OF 3 NUMBERS:

Step 1: Start

Step 2 : Read three numbers A, B & C

Step 3 : If $A > B$, then go to step 6

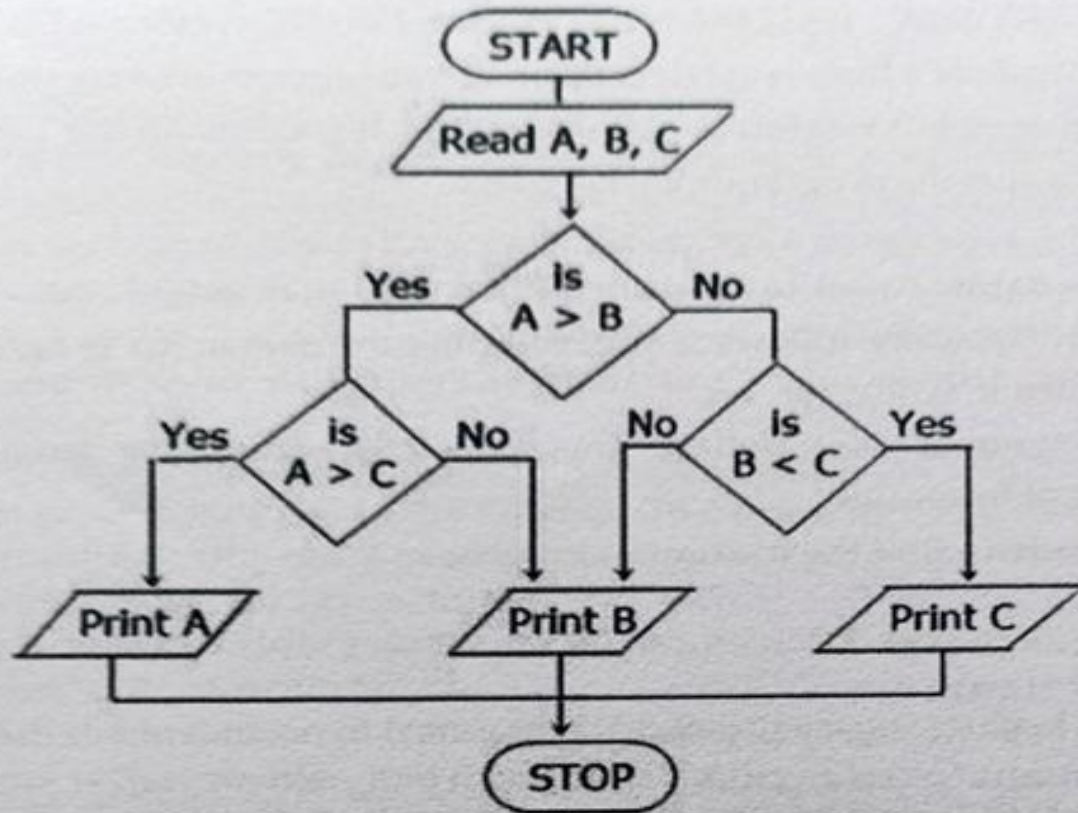
Step 4 : If $B > C$, then print B & go to step 8

Step 5 : print C is greatest & go to step 8

Step 6 : If $A > C$, then print A is greatest & go to step 8

Step 7 : Print C is greatest

Step 8 : End



PSEUDOCODE :

Definition:

- A Pseudocode is defined as a step-by-step description of an algorithm.
- Pseudocode does not use any programming language in its representation instead it uses the simple English language text as it is meant for human understanding rather than machine reading.
- It focuses on the logic of the algorithm rather than the syntax of a specific programming language.

Characteristics:

- Human-readable: Easier to read and understand than actual code.
- Structured: Typically follows a logical structure similar to programming languages, using keywords like if, while, for, etc.

Example: To find Sum and average of Two numbers.

Begin

WRITE "Please enter two numbers to add"

READ num1

READ num2

Sum = num1 + num2

Avg = Sum/2

WRITE Sum, Avg

End

Example : Write pseudocode that will take a number as input and tells whether a number is positive, negative or zero.

Begin

WRITE "Enter a number"

READ num

IF num > 0

THEN

WRITE "The number is positive"

ELSE IF num = 0

THEN

WRITE "The number is zero"

ELSE WRITE "The number is negative"

ENDIF

ENDIF

End

Example: Pseudocode to find the maximum number in a list:

```
FUNCTION findMax (list)
    SET max TO list [0]
    FOR each element in list DO
        IF element > max THEN
            SET max TO element
        ENDIF
    ENDFOR
    RETURN max
END FUNCTION
```

Algorithm : A precise sequence of steps to solve a problem.

Flowchart: A visual representation of the algorithm, showing the flow of control.

Pseudocode : A high-level, readable version of an algorithm that resembles programming code but written in natural language.

These tools are valuable for software developers to plan and communicate their ideas effectively before implementing the actual code.
